

EDUC III: Principles and Methods of Teaching

DESIGNING THE CURRICULUM AND INSTRUCTIONAL PLAN:

Processes, Considerations, and Frameworks for Effective Instruction Delivery

UNIVERSAL DESIGN FOR LEARNING (UDL)

This section serves as a comprehensive guide to the Universal Design for Learning (UDL) framework. It examines the historical, psychological, and neuroscientific foundations that empower educators to move beyond “one-size-fits-all” instruction toward flexible learning environments designed around the natural variability of learners.

I. DEFINITION:

According to Center for Applied Special Technology (CAST), Universal Design for Learning (UDL) is “a framework to improve and optimize teaching and learning for all people based on scientific insights into how humans learn. CAST further explains that UDL is an educational framework that guides the design of learning goals, assessments, methods, and materials to accommodate learner variability and reduce barriers to learning.

II. THE CORE PHILOSOPHY: SHIFTING THE DEFICIT MINDSET

At the center of UDL is a radical re-examination of what it means when a student struggles in school. Traditional pedagogy often operates under a medical or deficit model, viewing a struggling learner as an isolated issue to be remediated, separated, or fixed. UDL flips this dynamic completely, establishing that environmental rigidity is the true barrier to education.

The Fundamental Metaphor of UDL:

"If a plant does not grow, change the soil, not the plant."

The Plant: Represents student variability (differences in culture, language, neurotype, and life experience). This variability is natural, organic, and fixed.

The Soil: Represents the instructional design, curriculum, materials, assessments, and physical/digital learning environment. This is entirely flexible and under the teacher's control.

Instructional Takeaway: When a student is locked out of learning, UDL directs us to audit and enrich the environment (the soil) rather than pathologizing or labeling the student (the plant). In a UDL-aligned classroom, **the curriculum and the learning environment is redesigned to reduce barriers** with meaningful ways to access and demonstrate learning.

III. HISTORICAL EVOLUTION OF UDL

The framework used today is the result of decades of cross-disciplinary convergence, moving from physical engineering to advanced cognitive sciences, and ultimately expanding into modern equity-driven educational frameworks.

1. Architectural Blueprint (1970s)

Architect Ron Mace, a wheelchair user, coins "Universal Design" (UD). Rather than retrofitting clumsy ramps onto finished buildings, Mace designs physical spaces (like automatic sliding doors and sidewalk curb cuts) from the beginning. This highlights the "Curb-Cut Effect": what is essential for some (wheelchair ramps) turns out to be highly beneficial for all (parents with strollers, travelers with bags, delivery workers).

2. Radical Realization (1984)

Dr. David Rose (neuropsychologist) and Dr. Anne Meyer (educational psychologist) found the **Center for Applied Special Technology (CAST)** in Boston, initially trying to use digital technology to aid students with learning disabilities. They reach a radical realization: no matter how much assistive tech they give to a student, school textbooks and rigid teaching styles remain blockages. The student is not disabled; the rigid curriculum is.

3. Learning Brain (1990s)

The rise of rapid brain imaging technology (fMRIs) allows researchers to watch the learning brain in real-time. CAST maps these cognitive processes into three primary brain networks: *Affective (The "Why")*, *Recognition (The "What")*, and *Strategic (The "How")*. These networks serve as the **structural pillars of the core UDL guidelines**.

4. Equity Shift (Present)

UDL expands globally, integrated into policies (including inclusive K-12 localization programs). As classrooms grow culturally and linguistically diverse, the newly released UDL Guidelines 3.0 shift focus from merely "accommodating disabilities" to actively identifying and dismantling systemic classroom biases, honoring identities, and building true learner agency.

IV. PSYCHOLOGICAL & PHILOSOPHICAL UNDERPINNINGS

UDL is built upon well-established, validated educational and cognitive psychology theories:

- 1. Vygotsky's Zone of Proximal Development (ZPD) & Scaffolding:** Learning happens in the space between what a student can do independently and what they cannot do at all. *UDL directly operationalizes Vygotsky's theory by embedding flexible, temporary scaffolds (graphic organizers, vocabulary guides, models) into the environment, systematically fading these supports as the student builds internal capacity.*

2. **Cognitive Load Theory (Sweller):** Working memory possesses a rigid, finite capacity. When instructional design forces a student to waste mental energy dealing with unnecessary friction (such as deciphering a blurry visual, copying notes at high speed, or translating archaic accents), extraneous cognitive load consumes their capacity, leaving no room for actual conceptual processing. *UDL minimizes this load through multi-modal presentation.*
3. **Social Constructivism (Dewey & Vygotsky):** Learners are not passive "empty vessels" to be filled with lectures. Instead, they actively construct knowledge based on their unique prior experiences, culture, and social interactions. *UDL treats student variability as a major classroom resource rather than a liability.*
4. **Pragmatism & Progressive Education (John Dewey):** Dewey championed active "learning by doing" and democratic classrooms that accommodate the whole child. *UDL implements this directly by replacing passive, rigid factory-model compliance with real student voice, structured options, and personal accountability.*

V. EDUCATIONAL NEUROSCIENCE: HOW THE BRAIN LEARNS

UDL rejects the myth of the "average brain." Neuroscience proves that brain networks are as unique to an individual as their fingerprint. The UDL framework draws on an understanding of learning as involving interconnected affective, recognition, and strategic processes, which provide a useful conceptual foundation for designing multiple pathways for learning.

1. The Affective Network: The "Engine" (*The "Why" of Learning*)

Anatomy: Located deep in the core of the brain (the limbic system, including the amygdala and hippocampus).

Function: It monitors the environment for safety, emotion, interest, and motivation. It decides whether something is worth paying attention to or if it represents a threat.

Educational implication: Learners may differ in their interests, motivations, emotional responses, and capacity to sustain engagement. Factors such as anxiety, perceived threat, lack of belonging, or low personal relevance may create barriers to participation and learning.

2. The Recognition Network: The "Sensor" (*The "What" of Learning*)

Anatomy: Located in the back of the brain (the occipital, temporal, and parietal lobes).

Function: The recognition network helps learners take in, identify, and make meaning from information. It processes patterns, shapes, sounds, language, symbols, and other forms of sensory input. In simple terms, it helps answer: "What am I looking at, hearing, or experiencing?"

Educational implication: Learners may differ in how they perceive, process, and interpret information. When information is presented through a single modality or in a form that creates unnecessary barriers, some learners may have difficulty accessing or constructing meaning.

3. The Strategic Network: The "Pilot" (*The "How" of Learning*)

Anatomy: Located in the front of the brain (the prefrontal cortex).

Function: This is the executive control center. It is responsible for planning, organizing, executing physical actions, expressing ideas, and monitoring progress.

Educational implication: Learners may differ in their ability to plan, organize, coordinate actions, communicate ideas, and monitor their performance. Complex tasks may create barriers when learners must simultaneously manage several cognitive, linguistic, motor, or organizational demands.

Taken together, the three networks illustrate that learning is not a uniform process and that learner variability is an expected feature of educational contexts. Learners may differ in how they engage with learning, perceive and interpret information, and plan and demonstrate their understanding. Consequently, instructional barriers may arise when learning environments rely on a single means of engagement, representation, or action and expression. This understanding provides the neuroscientific and learning-science foundation for UDL, but it does not by itself prescribe specific instructional practices. The UDL Guidelines provide the framework through which these principles can be translated into intentional decisions about instructional design.

VI. THE DETAILED UDL GUIDELINES (VERSION 3.0)

Understanding how the brain learns gives us an important foundation for understanding why learners need different pathways to learning. The Affective, Recognition, and Strategic Networks show that learning involves different but interconnected processes: learners need to be engaged, they need accessible ways to perceive and understand information, and they need opportunities to act, practice, and demonstrate what they know.

These insights have significant implications for instructional design. If learners differ in how they engage with learning, access and process information, and demonstrate their understanding, then instructional environments must be intentionally designed to accommodate such variability. This principle is central to Universal Design for Learning (UDL).

“If learners have different ways of engaging, processing information, and demonstrating understanding, how should we design learning environments to account for that variability from the beginning?”

The UDL guidelines translate these principles into concrete considerations that teachers can use when designing goals, materials, methods, assessments, and learning environments.

Universal Design for Learning Guidelines

The goal of UDL is [learner agency](#) that is purposeful & reflective, resourceful & authentic, strategic & action-oriented.



Photo source:: CAST (2024). *Universal Design for Learning Guidelines version 3.0 [graphic organizer]*. Lynnfield, MA: Author.udlguidelines.cast.org

The UDL Guidelines 3.0 are organized through two intersecting dimensions: vertical organization according to the three UDL principles and horizontal organization across three rows: Access, Support, and Executive Function. This structure enables the Guidelines to be read both according to the different dimensions of learning and according to the types of support that can be designed within a learning environment.

Vertically, it is organized around the three UDL principles:

1. **Design Multiple Means of Engagement** – addresses how learners become involved and remain invested in learning
2. **Design Multiple Means of Representation** – representation addresses how learners access and construct meaning from information
3. **Design Multiple Means of Action & Expression** – addresses how learners interact with the learning environment and demonstrate their knowledge and understanding

These principles reflect different dimensions of the learning process.

CAST emphasizes that no single means of engagement, representation, or action and expression is optimal for all learners; therefore, providing appropriate options is central to UDL.

Horizontally, it is organized into three rows:

1. **Access, Support, and Executive Function** – focuses on increasing access to learning by providing options for Welcoming Interests & Identities, Perception, and Interaction.
2. **Support row** – focuses on supporting the learning process through Sustaining Effort & Persistence, Language & Symbols, and Expression & Communication.
3. **The Executive Function row** – addresses learners' capacity to regulate, organize, plan, and monitor their learning through Emotional Capacity, Building Knowledge, and Strategy Development.

The intersection of these two dimensions results in nine UDL Guidelines, with each guideline further elaborated through specific considerations that provide more detailed suggestions for instructional design. Thus, the framework can be understood as a progression from broad design principles to increasingly specific considerations for practice. CAST identifies **9 Guidelines** and **36 associated considerations** in Version 3.0

For example, under Design Multiple Means of Representation, the three guidelines are Perception, Language & Symbols, and Building Knowledge. These are further elaborated through considerations such as supporting multiple ways of perceiving information, clarifying vocabulary and language structures, connecting prior knowledge to new learning, and cultivating multiple ways of knowing and making meaning.

Therefore, the UDL Guidelines 3.0 should not be interpreted as a checklist requiring teachers to implement every option for every learner. Rather, they function as a flexible design framework that supports educators in identifying potential barriers and selecting appropriate options according to the learning goal, instructional context, and variability of learners. The Guidelines are intended to inform the proactive design of learning experiences rather than prescribe a single method of teaching.

VII. Classroom Applications of the UDL Guidelines

1. Design Multiple Means of Engagement

Guideline 7: Welcoming Interests & Identities

Consideration: Optimize choice and autonomy

In an English speaking lesson, rather than requiring all learners to discuss the same topic, the teacher may provide several topic options aligned with the same speaking competency. Students may select a topic that reflects their interests or experiences while still meeting the same learning objective.

2. Design Multiple Means of Representation

Guideline 1: Perception

Consideration: Support multiple ways to perceive information

During a lesson on literary analysis, the teacher may combine an oral explanation with a written excerpt, visual organizer, and highlighted textual evidence. This allows learners to access important information through complementary forms of representation.

3. Design Multiple Means of Action & Expression

Guideline 4: Interaction

Consideration: Vary and honor the methods for response, navigation, and movement

When assessing students' understanding of a historical issue, the teacher may allow learners to demonstrate their understanding through a written response, oral presentation, infographic, or recorded explanation, provided that the chosen format does not undermine the intended learning objective.

VIII. SUMMARY:

The UDL Guidelines are intended to function as a flexible framework for instructional design rather than an exhaustive checklist. Educators are not expected to implement every guideline or consideration in every learning experience. Instead, the framework supports educators in anticipating and identifying potential barriers within the learning environment and providing appropriate options that increase access, participation, and meaningful learning for learner variability.

Core Principles of Effective Instruction Design

This section provides a comprehensive guide of the core principles of effective instruction, drawing on research from cognitive science, studies of effective teachers, and research on instructional supports for complex learning tasks. It translates these research foundations into practical principles for designing and delivering systematic instruction.

I. Introduction

Teaching is both an art and a science. While creativity and passion are essential qualities of effective teachers, decades of educational research demonstrate that successful instruction also depends on the use of evidence-based teaching practices. Effective instruction does not happen by chance; rather, it results from careful planning, purposeful implementation, continuous assessment, and thoughtful reflection. Teachers who understand how students learn are better equipped to design lessons that promote meaningful learning, long-term retention, and higher academic achievement.

One of the most influential research-based frameworks for effective teaching is Barak Rosenshine's Principles of Instruction. These principles were developed through a synthesis of findings from three complementary bodies of research: cognitive science, research on highly effective teachers, and research on instructional supports (cognitive supports). Instead of proposing a new teaching theory, Rosenshine identified the instructional practices that consistently appear in classrooms where students achieve high levels of learning. These practices have since become widely recognized as practical guidelines for designing effective instruction across grade levels and subject areas.

Although the principles are straightforward, they are grounded in a sophisticated understanding of how people learn. They recognize that students have limited working memory, require opportunities for active practice and research-supported models available to classroom teachers today.

II. Research Foundations of Effective Instruction

Before exploring the instructional principles themselves, it is important to understand the research that supports them. Rosenshine organized his framework around three complementary sources of evidence.

1. Research in Cognitive Science

Cognitive science investigates how the human mind acquires, processes, stores, and retrieves information. One of its most important findings is that working memory is limited. Learners can process only a small amount of new information at a time, and presenting too much content simultaneously can overwhelm them. Consequently, effective instruction should present information in manageable segments, connect new ideas with prior knowledge, and provide opportunities for repeated practice to strengthen long-term memory.

2. Research on Master Teachers

Studies comparing highly effective teachers with less effective teachers revealed consistent instructional practices that contribute to greater student achievement. Successful teachers devote more time to explaining concepts, modeling procedures, asking questions, checking for understanding, guiding student practice, and providing immediate corrective feedback. They carefully monitor student progress and ensure that learners achieve mastery before introducing new material.

3. Research on Cognitive Supports

Instructional supports such as modeling, scaffolding, worked examples, and think-aloud strategies help students perform tasks that they could not yet complete independently. These temporary supports reduce cognitive load while learners develop competence. As students become more proficient, the supports are gradually withdrawn, enabling learners to work independently. Students benefit from explicit instruction, and need continuous feedback before they can perform tasks independently. For this reason, Rosenshine's framework remains one of the most practical and research-supported models available to classroom teachers today.

III. The Ten Principles of Instruction

1. Begin a Lesson with a Short Review of Previous Learning

Effective instruction begins by briefly revisiting previously learned knowledge and skills. Review can involve questions, short practice activities, correcting previous errors, or recalling important facts and procedures.

The purpose is not simply to “warm up” students. Review helps make relevant prior knowledge readily available for the new lesson and strengthens learning over time. Rosenshine recommends beginning lessons with a short review of previous learning, including review of vocabulary, facts, concepts, and previously learned skills.

For example, before introducing the new lesson on Subject-Verb Agreement, the teacher may ask:

- “What is a subject?”
- “What is a verb?”

- “Which verb agrees with she: eat or eats?”

Students answer several quick questions before the teacher introduces more complex cases.

The teacher should:

- Use short retrieval questions.
- Review prerequisite knowledge.
- Correct misconceptions immediately.
- Connect the review to today's objective.

2. Present New Material in Small Steps with Student Practice After Each Step

Teachers should avoid presenting large amounts of unfamiliar information at once. Instead, new material should be divided into small, manageable steps, with opportunities for students to practice after each step. This helps prevent students from becoming overwhelmed and allows teachers to identify difficulties before moving forward.

For instance, instead of teaching all aspects of solving quadratic equations in one explanation, the teacher proceeds:

- Step 1: Identify the quadratic equation.
- Step 2: Factor the expression.
- Step 3: Set each factor equal to zero.
- Step 4: Solve for the unknown.

After each step, students complete a short practice item.

The teacher should:

- Break complex procedures into steps.
- Present one new element at a time.
- Practice immediately after introducing each step.
- Check understanding before proceeding.

3. Ask a Large Number of Questions and Check the Responses of All Students

Questioning should not be limited to a few students who volunteer answers. Effective instruction provides frequent opportunities for all students to think and respond. Questioning allows students to actively process information while giving the teacher immediate information about their understanding.

For instance, after explaining evaporation, instead of asking, “Does everyone understand?”, the teacher asks: “What happens to water molecules when they gain energy?”

Students respond using:

- mini-whiteboards,

- response cards,
- individual answers,
- pair discussions followed by responses.

The teacher can quickly see who understands and who needs additional explanation.

The teacher should:

- Ask frequent questions.
- Give students time to think.
- Randomly select students when appropriate.
- Use whole-class response techniques.
- Check responses rather than assuming understanding.

4. Provide Models

Teachers should demonstrate how to perform tasks or solve problems before asking students to work independently. Modeling, including think-alouds and worked examples, helps students understand both the process and the reasoning behind successful performance.

Models can include:

- worked examples
- demonstrations
- completed samples
- think-alouds
- teacher modeling of procedures

Instead of telling students, “Write a good argumentative paragraph.” the teacher displays a model paragraph and thinks aloud:

“First, I need to state my claim. Then I need evidence that supports the claim. Finally, I need to explain how the evidence supports my argument.”

The teacher then highlights each component.

The teacher should:

- Show an exemplar.
- Demonstrate the process.
- Think aloud.
- Explain decision-making.
- Point out common errors.

Rosenshine identifies modeling as an important instructional support, particularly when students are learning complex tasks.

5. Guide Student Practice

After demonstrating a new skill, teachers should provide guided practice where students apply their learning with support. During this stage, teachers offer prompts, feedback, and corrections to ensure that students develop accurate understanding before working independently.

The teacher first models:

“In my opinion, school uniforms are beneficial because...”

Students then practice the same structure with a partner.

The teacher circulates, listens, prompts students when necessary, and corrects recurring errors.

Only after sufficient guided practice are students asked to deliver their own response independently.

The teacher should

- Practice together before assigning independent work.
- Monitor student responses.
- Provide prompts and cues.
- Correct errors.
- Gradually reduce assistance.

6. Check for Student Understanding

Teachers should regularly assess students' understanding throughout instruction rather than waiting until the end of the lesson. By asking students to explain concepts, summarize ideas, or demonstrate procedures, teachers can identify misunderstandings early and adjust instruction accordingly.

To illustrate, after explaining the causes of the Philippine Revolution, the teacher asks students to:

“Identify one political, one economic, and one social factor and explain the relationship between them.”

The teacher examines responses and notices that many students confuse cause and effect. Instead of proceeding to the next topic, the teacher briefly reteaches the distinction.

The teacher should:

- Ask students to explain rather than simply repeat.
- Use formative assessment.
- Observe student performance.
- Identify misconceptions.
- Reteach when necessary.

7. Obtain a High Success Rate

Students learn best when they experience consistent success during instruction. Lessons should be challenging enough to promote growth but structured so that learners can correctly apply new knowledge with guidance. High success rates build confidence, motivation, and mastery while reducing the likelihood of reinforcing errors.

Suppose students are learning how to solve systems of equations.

If most students are repeatedly getting the problems wrong, the teacher should not simply assign more problems.

Instead, the teacher might:

- return to the worked example;
- model the difficult step again;
- provide a simpler problem;
- guide students through another example;
- gradually increase the difficulty.

The teacher should:

- Monitor error rates.
- Adjust task difficulty.
- Provide additional modeling.
- Give more guided practice.
- Avoid moving forward when students are consistently unsuccessful.

8. Provide Scaffolds for Difficult Tasks

Scaffolds are temporary instructional supports that help students accomplish challenging tasks. These supports such as graphic organizers, prompts, checklists, and sentence starters enable learners to succeed while gradually developing independence. As students become more competent, the scaffolds are systematically removed.

Scaffolds may include:

- prompts
- checklists
- graphic organizers
- cue cards
- partially completed examples
- sentence starters
- worked examples
- teacher modeling

The support should be gradually removed as students develop greater competence.

For example, instead of asking students immediately to write a three-page argumentative essay, the teacher provides:

- Step 1: Choose a position.
- Step 2: Complete a claim-evidence-reasoning organizer.
- Step 3: Develop a thesis.
- Step 4: Draft one body paragraph.
- Step 5: Develop a conclusion.

The organizer and sentence starters are eventually removed as students become more capable.

9. Require and Monitor Independent Practice

Independent practice allows students to reinforce newly acquired knowledge and develop fluency. During this stage, teachers continue to monitor students' work, provide feedback, and address errors to ensure that learners apply concepts correctly and confidently.

Rosenshine emphasizes that students need sufficient preparation before independent practice and that teachers should monitor this practice to ensure students are working successfully.

For example, after the teacher has explained → modeled → guided → checked understanding, students independently complete 10 sentences requiring them to apply the target grammatical structure. The teacher circulates and monitors rather than reteaching the entire lesson to individual students.

The teacher should:

- Assign appropriate independent practice.
- Monitor student performance.
- Look for recurring errors.
- Provide feedback.
- Use errors to inform subsequent instruction.

10. Engage Students in Weekly and Monthly Review

Learning is strengthened through regular review over time. Weekly and monthly review activities help students retrieve previously learned information, reinforce connections among concepts, and transfer knowledge into long-term memory. This continuous review promotes retention and supports future learning.

Rosenshine emphasizes the importance of weekly and monthly review, including review of facts, vocabulary, concepts, problem-solving skills, and previously learned material.

At the beginning of Friday's class, the teacher gives a five-minute retrieval activity covering:

- vocabulary from the previous week
- grammar learned earlier in the month
- a previously practiced speaking structure
- one reading strategy

The activity does not introduce new content. It helps students retrieve and reconnect previously learned knowledge.

The teacher should:

- Use cumulative review.
- Revisit important concepts.
- Mix old and new material.
- Use retrieval questions.
- Include previously learned skills in later activities.

A research-based summary of 17 instructional principles derived from the research discussed in Rosenshine's *Principles of Instruction*. The 17 principles overlap with and provide greater detail than the 10 principles used to organize the main article. They emphasize review of prior learning, small-step presentation, clear explanations, questioning, active and guided practice, modeling, worked examples, feedback, reteaching, and the gradual transition toward independent practice.

17 Principles of Effective Instruction

The following list of 17 principles emerges from the research discussed in the main article. It overlaps with, and offers slightly more detail than, the 10 principles used to organize that article.

- Begin a lesson with a short review of previous learning.
- Present new material in small steps with student practice after each step.
- Limit the amount of material students receive at one time.
- Give clear and detailed instructions and explanations.
- Ask a large number of questions and check for understanding.
- Provide a high level of active practice for all students.
- Guide students as they begin to practice.
- Think aloud and model steps.
- Provide models of worked-out problems.
- Ask students to explain what they have learned.
- Check the responses of all students.
- Provide systematic feedback and corrections.
- Use more time to provide explanations.
- Provide many examples.
- Reteach material when necessary.
- Prepare students for independent practice.
- Monitor students when they begin independent practice.

—B.R.

Note. The 17 principles emerge from the research discussed in Rosenshine's (2012) *Principles of Instruction* and provide greater detail than the 10 principles used to organize the article.

Photo source: Rosenshine, B. (2012). *Principles of instruction: Research-based strategies that all teachers should know*

IV. Summary

Rosenshine's Principles of Instruction presents a research-based framework for effective classroom instruction. Drawing from cognitive science, observations of effective teachers, and research on instructional supports for complex tasks, the framework emphasizes that learning is strengthened when teachers provide systematic support as students acquire and practice new knowledge and skills.

The ten principles follow a progression from activating prior knowledge and introducing new material, to modeling, questioning, guided practice, checking for understanding, scaffolding, and independent practice, followed by regular review. Effective instruction therefore involves more than presenting information; it requires teachers to manage the amount of new material, provide sufficient practice and feedback, monitor student learning, and gradually transfer responsibility to learners.

The principles should not be viewed as rigid steps that must be followed identically in every lesson. Rather, they provide research-informed instructional practices that teachers can adapt according to the content, learners, and instructional context.

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